

Lotka—Volterra modeling of a three species microbial community with respect to spatial complexity

Leander Kühne, BSc

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1 Abstract

This project implements a Lotka—Volterra model to analyse time series growth data, derive interaction coefficients and explore how a three-species interaction can be modeled from limited growth data of only two of the three species framing the third species as a perturbation to the interaction of the others. Fluorescently labeled *Pseudomonas putida* and *Bacillus subtilis* bacteria were co-cultured with a non-labeled *Trichoderma viride* fungus in micro-scale mazes with different degrees of fractal complexity. The R package **gauseR** (Mühlbauer et al. 2020) was employed to fit logistic growth curves to the experimental data and calculate interaction coefficients ($a_{ii}, a_{jj}, a_{ij}, a_{ji}$), growth rates (r) and carrying capacities (K).

2 Introduction

2.1 Importance of community and habitat complexity in microbial ecology

In microbial ecology, most experimental research is focused on either community response to a generalized change in ecosystem properties (e.g. temperature, pH, nutrient availability, toxins) by characterizing communities on a very broad and data generation driven “-omics” approach with limited reconnaissance for functional and mechanistic processes, or it focuses on two species interactions — of oftentimes synthetic consortia — with limited transferability to *in situ* complexity, neglecting the emergence of ecosystem properties and higher order interactions in interaction networks of complex communities .

Moreover, most of this culturing-technique based approaches is based entirely on well-mixed “on a shaker”-type conditions. This poses great challenges for the study of community driven effects with relevance of the spatial arrangement like biofilm formation, quorum sensing or niche differentiation (Orr et al. 2025). This is especially limiting for the study of soil habitats, since they are characterized by a very high spatial complexity and thus habitat diversity. On a side-note, the underrepresentation of community complexity and spatial structure might be part of the explanation for the uncultureability-problem that experimental microbiological research is facing since it’s very beginnings.

P. putida and *B. subtilis* are often co-isolated from soil samples and *T. viride* is also a common member of soil microbial communities. Therefore all three species are thought to be stably coexisting in soil habitats under natural conditions. The underlying question of these experiments specifically are how *T. viride* affects the growth of *P. putida* and *B. subtilis* facilitating or hindering coexistence, by suppression or support of one or both of the species or by altering the interactions between the two.

2.2 Old but gold: The simple yet powerful framework of the Lotka—Volterra equation

The Lotka-Volterra equation is while being a mathematically relatively simple differential equation still a very powerful approach in ecology that can account for very complex and multi-layered networks of interactions without the need to have a complete mechanistic understanding of these interactions on a molecular level. The canonical framework of

The basic Lotka—Volterra equation employed throughout this study describes the changes per unit time in the abundance of species N_i as

$$\frac{dN_i}{dt} = N_i(r_i + a_{ii}N_i + \sum_j (a_{ij}N_j)) \quad (1)$$

with r_i as the intrinsic growth rate, a_{ii} being the intra-specific competition (i.e. the effect of species i on itself, a_{ij} describing the effect of species j on species i and N_j as the abundance of species j .

With the approach to reflect the interaction dynamics of this three species system in a simple Lotka-Volterra model I am trying to approximate on the dynamics with a simple model approach to see whether experimental data can be fitted satisfyingly on this degree of complexity already or if more sophisticated models need to be employed. Also the attempt to model the data with a LV model can give insights if the interactions can be described by transient competition or if higher-order interactions are at work.

In this experimental setup time series growth data is only available for two of the three species. The overall aim is to see whether the experimental data can be properly represented by a Lotka-Volterra model and investigate if that approach is a viable method to narrow in on mechanistic insights to the interactions of this three species model in a spatially structured habitat.

Generalized LV:

$$\frac{dx_i(\mathbf{t})}{dt} = \mathbf{x}_i(\mathbf{t}) \left(\mathbf{b}_i + \sum_{j=1}^n \mathbf{a}_{ij}x_j(\mathbf{t}) \right) \quad i = 1, \dots, n$$

3 Methods and Data

3.1 Experimental methods

The wet lab experimental setup consists of a semi-continuous culturing experiment on microfluidic chips comprised with micro-scale mazes of different fractal complexity (f0-f5, Hilbert curve

space filling design; Hilbert (1935)) and normalized internal volume. The chips consist of a culture/media inoculation area from where mazes branch off Figure 1.

P. putida and *B. subtilis* have a genomically integrated expression cassette for green- and red-fluorescent protein (GFP/RFP) production respectively. Fluorescence intensity levels are used as a proxy for species abundance throughout this study.

Culturing experiments were carried out over 22 days with a 48 h media-replenishment cycle replicating close-to unlimited nutrient supply conditions (M9 minimal medium). In the same cycle the mazes were imaged by epi-fluorescence microscopy collecting spatially resolved GFP and RFP signals to approximate species abundance.

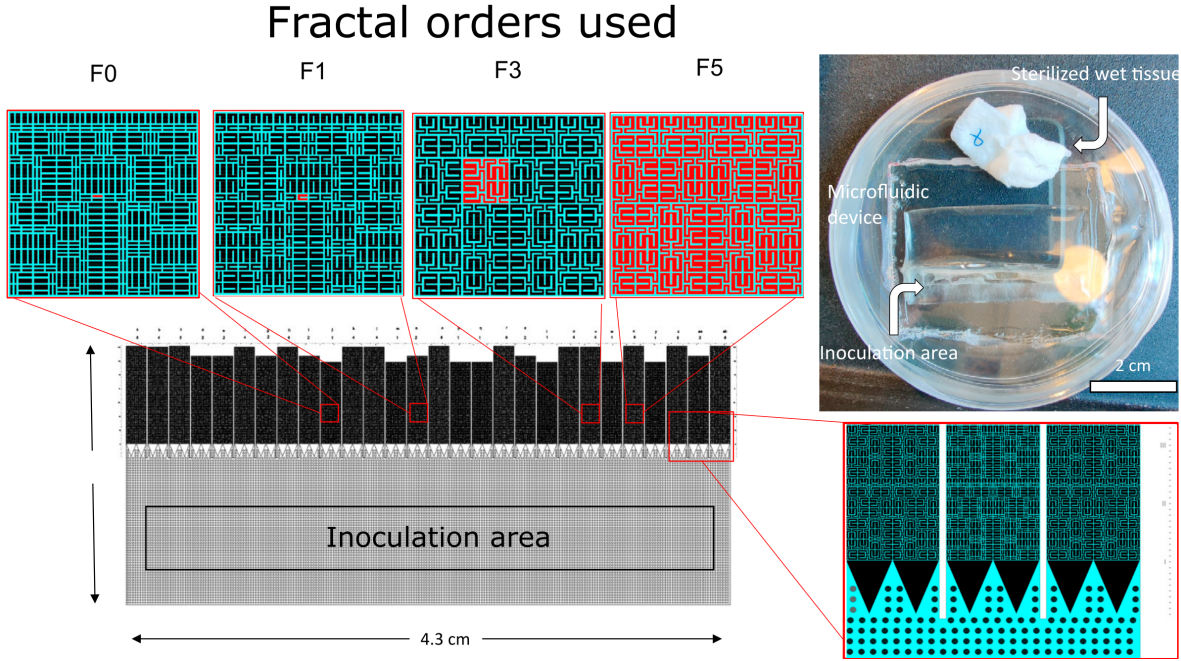


Figure 1: Fractal chip simulating microhabitats of different physical complexity. Figure kindly provided by Arellano-Caicedo et al. (2023).

3.2 Software and Code

The main analysis was carried out using R version 4.6.1. The Lotka—Volterra modeling was done using the **gauser** package version 1.3. Additionally the following software was used for data wrangling, processing and report compiling: ImageJ 1.54 (raw TIFF microscopy image data processing), RStudio 2026.09.0 Build 174 (main IDE), quarto 1.10.18 (report writing and rendering), pandoc 3.7.0.2 (report pdf rendering), LuaHBTeX 1.24.0 (TeX Live 2026, as pandoc compiler).

Additional R packages used apart from R base were: `tidyverse`, `ggh4x`, `kableExtra`, `readxl`.

3.3 Data analysis and modeling workflow

Raw microscopy images were aligned and processed by background subtraction using an ImageJ rolling ball algorithm with the rolling ball diameter matching the width of the maze channels.

- your workflows; BUT: usually no command lines

4 Results

Here begins the text describing the results

First figure:

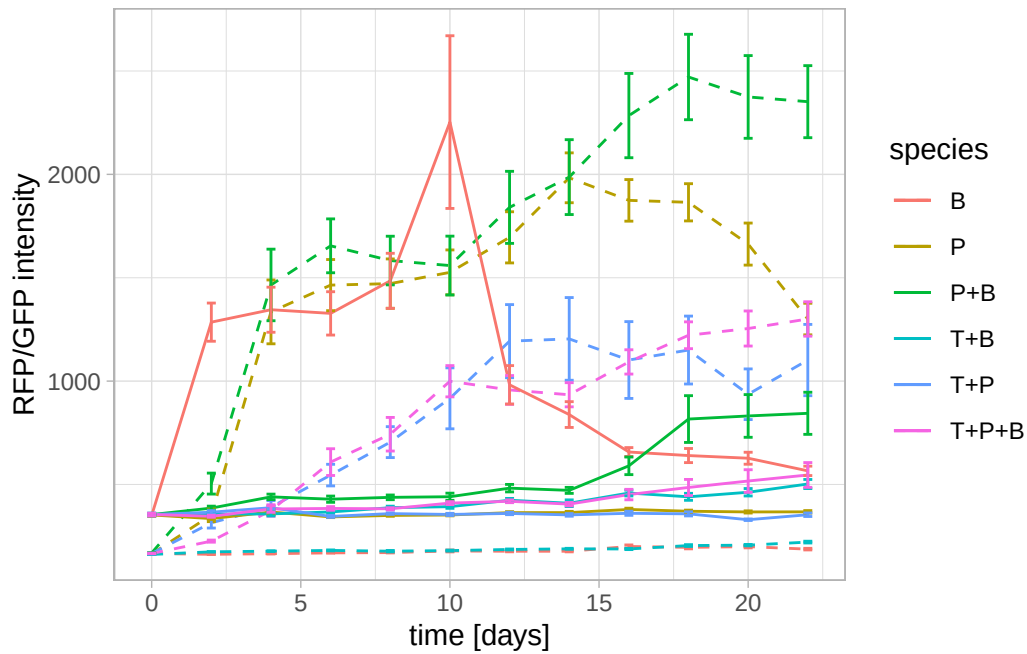


Figure 2: RFP (solid line) and GFP (dashed line) fluorescence intensity on microfluidic mazes with fractal order 5.

Second figure:

Heatmap of interaction coefficients

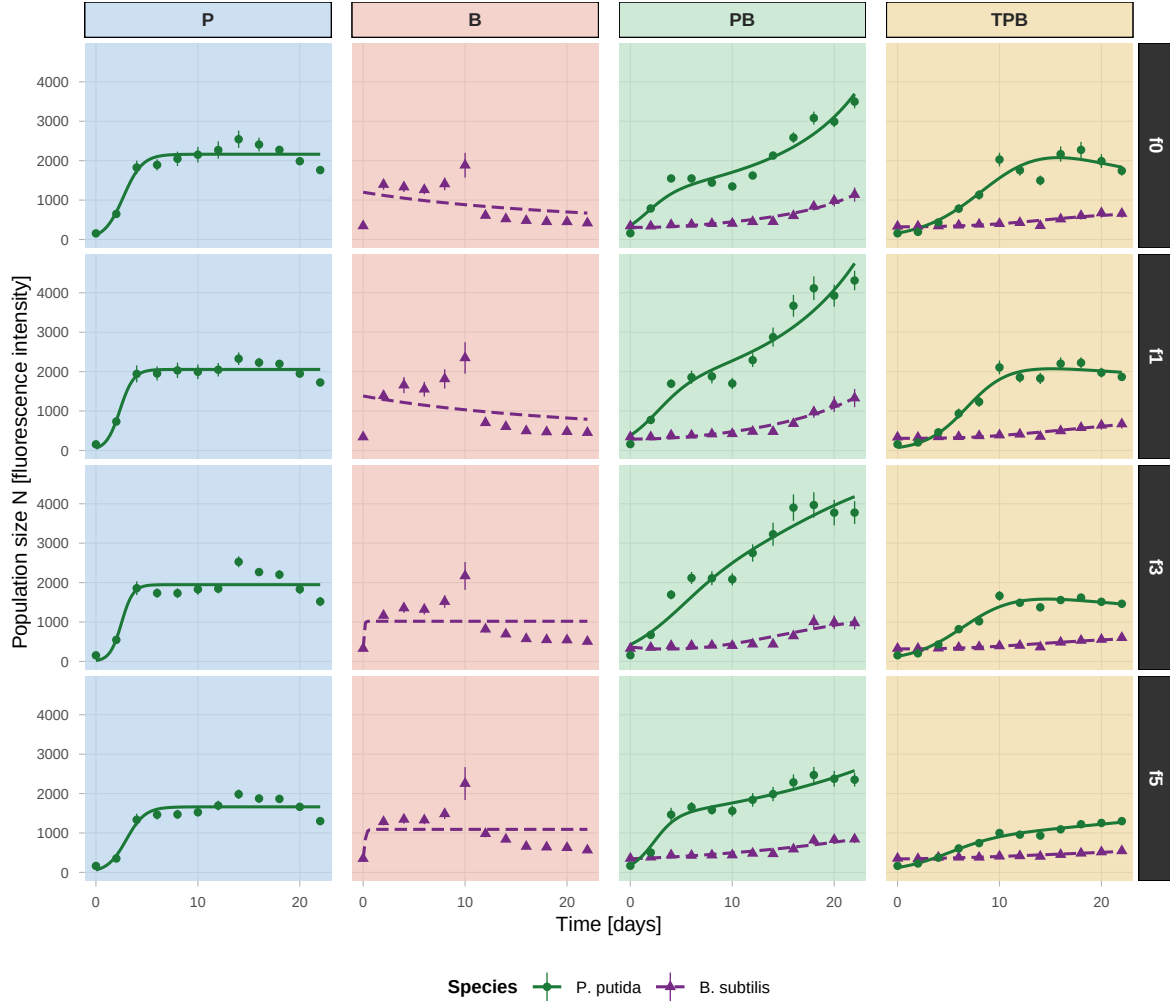


Figure 3: Observed (points, $\pm$ SE) and predicted (lines) growth curves across fractal levels (rows) and conditions (columns).

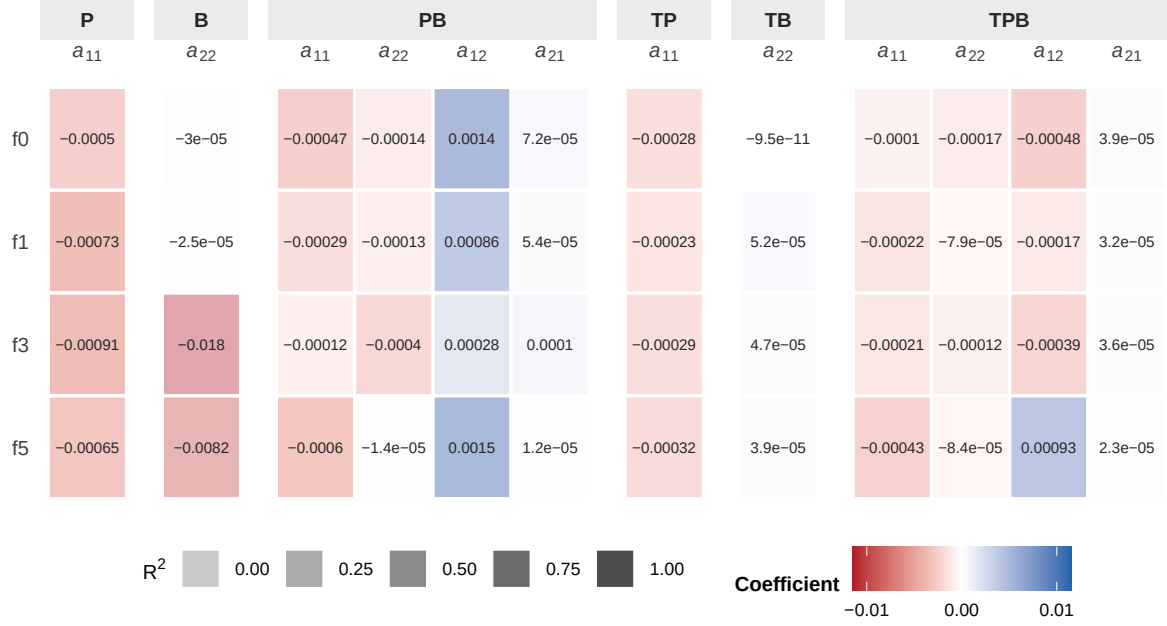


Figure 4: Interaction coefficients per fractal level (rows) and condition (columns). Tile colour shows the estimated coefficient (blue: positive, red: negative); colour saturation shows R^2 of the fit of the species whose growth equation contains the coefficient (faint = poor fit). Numbers are the coefficient values.

4.1 Lotka—Volterra modeling using the `gauseR` library

4.1.1 Parameterization and regression fit for *Pseudomonas* monoculture

Time-lagged data was generated using the `get_lag` function and the obtained data was used to calculate the per-capita growth rate.

The per-capita growth rate for the logistic growth model was defined as:

$$\frac{dN}{dt}/N = r - a_{ii}N$$

with r being the intrinsic growth rate, and a_{ii} the self-limitation in growth of the species.

The per-capita growth rate was plotted against the abundance and a linear regression was fitted to obtain the needed parameters r (read as the y-intercept) and K (read as the x-intercept). a_{ii} is the slope of the linear regression.

5 Discussion

- Discussion (did we succeed w.r.t. the project aims? did we make particular assumptions? do our methods have weaknesses? how do the results compare to previous expectations and previous work)

6 Statement of authorship, contributions and use of LLMs

This report is submitted as the practical project for the class “Practical Course in Modelling and Systems Biology” (2026S 300332-1). I originally wanted to conduct the modeling analyses with my own masters thesis data from co-culturing experiments with *P. putida*, *B. subtilis* and a complex soil bacterial community isolated from soil samples as a natural inoculum. However, since I was unable to have the necessary experimental data ready in time for the due date of this report I herein use the data of my colleague Maryam Al-Jalahwee, BSc who conducted similar experiments with *Trichoderma viride* as the third interaction entity. All data arranging, sorting and subsequent analysis has been conducted by the author.

I used CLAUDE *Sonnet 5* and OpenAI GPT-5.6 *Luna* for generation and improvements of R code. In particular I used it for the building of functions to automate my data processing pipeline for all the different conditions and fractal levels I have in my data, for bug fixing, efficiency improvements on the code I provided and for configuring layout, design and arrangement of plots and tables. I also used it to help me better understand the functionality of the `gauseR::test_goodness_of_fit` functionality.

7 References

- Arellano-Caicedo, Carlos, Pelle Ohlsson, Martin Bengtsson, Jason P. Beech, and Edith C. Hammer. 2023. “Habitat Complexity Affects Microbial Growth in Fractal Maze.” *Current Biology* 33 (8): 1448–1458.e4. <https://doi.org/10.1016/j.cub.2023.02.064>.
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8 Supplementary Material

The complete code that contains all data handling and also compiles this report reproducibly can be found as a Quarto document [GitHub](#).

Table 1: Estimated Lotka-Volterra interaction coefficients for cocultures of *P. putida* (1) and *B. subtilis* (2) on microfluidic chips with mazes of increasing fractal complexity (f0-f5). a_{11} and a_{22} are the intra-specific effects of *P. putida* and *B. subtilis*, respectively; a_{12} and a_{21} are the inter-specific effects of *B. subtilis* on *P. putida* and vice versa. Negative coefficients indicate competition/inhibition, positive coefficients promotion/enhancement.

Fractal	<i>P. putida</i>		<i>B. subtilis</i>		<i>P. putida</i> + <i>B. subtilis</i>			<i>T (name?)</i> + <i>P. putida</i>			<i>T (name?)</i> + <i>B. subtilis</i>			<i>T (name?)</i> + <i>P. putida</i> + <i>B. subtilis</i>		
	a_{11}	a_{22}	a_{11}	a_{22}	a_{11}	a_{12}	a_{21}	a_{11}	a_{12}	a_{21}	a_{22}	a_{11}	a_{12}	a_{21}	a_{22}	a_{21}
f0	-5e-04	0.0000	-5e-04	-1e-04	-5e-04	0.0014	1e-04	-3e-04	-1e-04	0e+00	0e+00	-1e-04	-2e-04	-5e-04	0	0
f1	-7e-04	0.0000	-3e-04	-1e-04	-2e-04	0.0009	1e-04	-2e-04	-1e-04	1e-04	1e-04	-2e-04	-1e-04	-2e-04	0	0
f3	-9e-04	-0.0181	-1e-04	-4e-04	-3e-04	0.0003	1e-04	-3e-04	-1e-04	0e+00	0e+00	-2e-04	-1e-04	-4e-04	0	0
f5	-7e-04	-0.0082	-6e-04	0e+00	-3e-04	0.0015	0e+00	-3e-04	-1e-04	0e+00	0e+00	-4e-04	-1e-04	9e-04	0	0

Table 2: Goodness of fit of the fitted growth curves against the observed means, per species. $R^2 = 1 - SS_{\text{res}}/SS_{\text{tot}}$ (can be negative for fits worse than the mean); RMSE in the units of N (fluorescence intensity). Model predictions were interpolated at the observed time points.

Condition	Fractal	<i>P. putida</i>		<i>B. subtilis</i>	
		R^2	RMSE	R^2	RMSE
<i>P. putida</i>	f0	0.913	201.964	—	—
	f1	0.942	150.121	—	—
	f3	0.836	263.606	—	—
	f5	0.885	186.036	—	—
<i>B. subtilis</i>	f0	—	—	0.169	466.131
	f1	—	—	0.137	608.991
	f3	—	—	0.133	487.614
	f5	—	—	0.161	467.490
<i>P. putida</i> + <i>B. subtilis</i>	f0	0.935	240.730	0.956	55.194
	f1	0.929	343.371	0.949	75.043
	f3	0.930	323.308	0.861	95.776
	f5	0.954	148.544	0.880	60.509
<i>T (name?)</i> + <i>P. putida</i>	f0	0.966	99.799	—	—
	f1	0.969	124.024	—	—
	f3	0.942	126.850	—	—
	f5	0.936	90.757	—	—
<i>T (name?)</i> + <i>B. subtilis</i>	f0	—	—	0.845	22.368
	f1	—	—	0.951	13.095
	f3	—	—	0.956	10.777
	f5	—	—	0.943	11.091
<i>T (name?)</i> + <i>P. putida</i> + <i>B. subtilis</i>	f0	0.909	225.997	0.865	45.482
	f1	0.960	151.623	0.896	38.554
	f3	0.959	110.466	0.904	28.892
	f5	0.979	54.865	0.933	15.762